Over Current Grading

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Abstract— **Relay coordination is very important in power system protection. Relays are devices that protect the machines, stations, and feeders in power system. They are devices that detect and send signals to circuit breakers which is the device used for opening circuit. Relays are considered the brain of the power system protection and should be coordinated carefully and accurately. This report will be handling over current relay and its types, also will have a calculations for a circuit and coordinating its relays.**

I. INTRODUCTION

Power system protection is a field of electrical power engineering concerned with the protection of electrical power systems against failures by isolating defective areas from the rest of the network. A protection scheme's goal is to maintain the power system stable by isolating only the components that are failing while keeping as much of the network as feasible operational. Protection devices are the equipment that are used to protect power systems from problems. In most cases, protection systems are made up of five parts:

- Current and voltage transformers are used to reduce the electrical power system's high voltages and currents to levels that relays can handle.
- Circuit breakers are ordered to open/close the system depending on relay and auto recloser commands.
- Protective relays detect the fault and trigger a trip, or disconnection.
- Batteries to provide electricity in the event of a power outage in the system.
- Communication channels for analysing current and voltage at line's distant terminals and tripping equipment from afar.

- Fuses are capable of both sensing and disconnecting problems in portions of a distribution system.

Insulation failure, fallen or broken transmission lines, faulty circuit breaker activation, short circuits, and open circuits are all possible failures. The goal of installing protection devices is to protect assets while also providing a steady supply of electricity. Electrical disconnect switches, fuses, and circuit breakers make up switchgear, which is used to control, protect, and isolate electrical equipment. Under normal load current, switches are safe to open (however some switches are not safe to function under normal or abnormal conditions), whereas protective devices are safe to open under fault current. Protective systems for critical equipment may be entirely redundant and independent, whereas protection for a minor branch distribution line may be relatively simple and low-cost. Mainly, this report is going to discuss the importance of over current relay which is a type of relays that detect over current in the circuit.

A relay is a switch that is controlled by electricity. A set of input terminals for a single or multiple control signals, as well as a set of functioning contact terminals, make up the device. The switch can have any number of contacts in any contact form, including make contacts, break contacts, and combinations of the two.

Relays are employed when an independent low-power signal is required to control a circuit, or when multiple circuits must be controlled by a single signal. Relays were first utilised as signal repeaters in long-distance telegraph circuits, refreshing the signal coming in from one circuit by transmitting it on another. Relays were widely employed to conduct logical operations in telephone exchanges and early computers. The overcurrent relay is defined as a relay that only activates when the current value exceeds the relay setting time. It protects the power system's equipment against fault current.

II. OVER CURRENT RELAYS

There would essentially be a current coil in an over current relay. When normal current flows through the coil, the magnetic effect generated by the coil is insufficient to move the relay's moving element, because the restraining force is larger than the deflecting force in this circumstance. The magnetic effect rises as the current through the coil increases, and beyond a certain level of current, the deflecting force generated by the coil's magnetic effect crosses the restraining force. As a result, the moving element in the relay begins to move, changing the contact position. Although there are many various varieties of overcurrent relays, the essential functioning principle is the same for all of them.

III. TYPES OVER CURRENT RELAYS

A) *Inverse over current relay:*

The operation time of these relays is inversely proportional to the current. As a result, high current overcurrent relays will work faster than lower current ones. There are three forms of inverse: standard, very inverse, and extremely inverse. The relay operation time is inversely proportional to the fault current when measured by both time and current. The 'time dial setting' of an overcurrent relay can be adjusted to increase (slow down) the operational time. The fastest working time (lowest time dial setting) is usually 0.5, and the slowest is 10 as shown in figure 1.

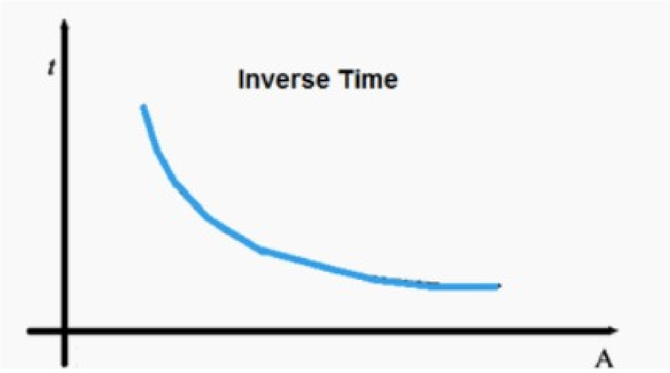

Figure 1: inverse over current relay curve

The operation of this type of relay does not have an intentional time delay. The contacts of the relay are immediately closed if the current inside the relay exceeds the operational value. The duration between closing contacts and instantaneous pickup value is quite short. This relay's main feature is its speed of operation, since it operates quickly when the impedance between the relay and the source is less than the value specified in the selection. They're utilised to keep systems safe against circulation currents and earth faults and for distribution lines. So, to sum up this type of relay of relay operates only when the current exceeds the pick-up value of the relay, and its operating time mainly depends on the magnitude of the fault current. The inverse over current relay gives inverse time characteristics at lower value of fault current, but it gives a definite time characteristic at higher values. And inverse time current characteristics is obtained when the plug setting multiplier value is below 10, but for the values between 10 and 20, the performance of the relay tens to be a definite time characteristic.

B) *Instantaneous time over current relay:*

nstantaneous over current relay operates when the current reaches a predefined value, instantaneous over current relay activates instantly. When the amount of the operating current is inversely proportional to the magnitude of the energize quantity, this sort of relay operates. The operational time of the relay is reduced when the current is increased. The relay's operation is dependent on the current's magnitude. When the current value is less than the pickup current, the relay will not work, as shown in figure 2.

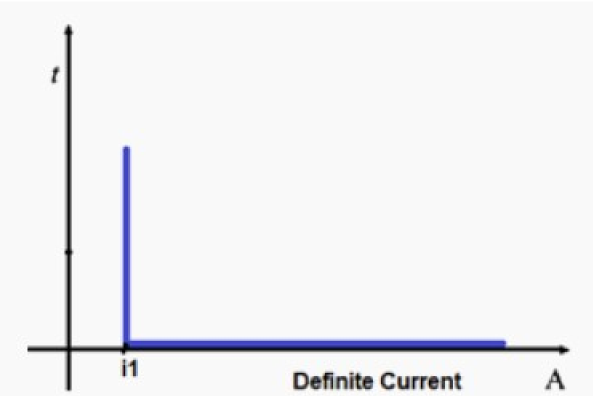

Figure 2: instantaneous over current curve

nverse time relays come in three different varieties and are used to protect distribution lines. In most cases, a current coil is being used to wound a magnetic core. When here isn't enough current in the coil, the NO contacts in he relay are held open by a piece of iron fitted with a inge support and a restraining spring. When the coil's current reaches a certain level, the attractive force is trong enough to draw the iron piece towards the magnetic core, closing the no contacts. as shown in figure 3.

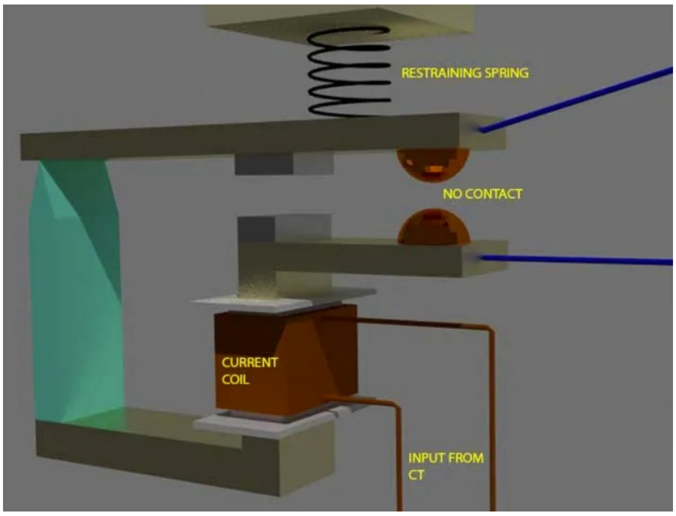

Figure 3: instantaneous over current relay components

The pre-set current in the relay coil is referred to as pickup setup current. This relay is known as an instantaneous over current relay because it acts as soon as the coil current exceeds the pick upsetting current. There is no time delay implemented on purpose. However, there is always a time delay that we cannot avoid. In fact, an instantaneous relay's operation time is on the order of a few milliseconds.

C) Definite time relay:

This relay is made by adding an intentional time delay after picking up the current value. A definite time overcurrent relay can be set to issue a trip output once it picks up for a specific period of time. As a result, it has a pickup adjustment and a time setting adjustment as shown in figure 4.

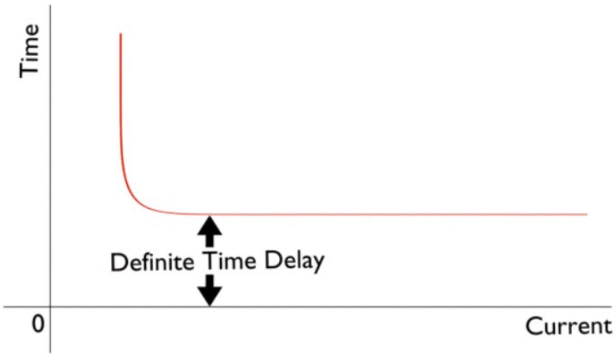

Figure 4: characteristics of definite time relay

D) Inverse Definite Minimum Time Over Current Relay:

In an overcurrent relay, ideal inverse time characteristics are impossible to acquire. The secondary current of the current transformer grows in proportion to the current in the system. The relay current coil receives the secondary current. When the CT gets saturated, however, an increase in system current does not result in a proportionate rise in CT secondary current. From this phenomenon, it is obvious that an inverse time relay exhibits a specific inverse feature from a trick value to a given range of defective level. However, after this level of fault, the CT becomes saturated, and the relay current does not grow with the system's defective level. There will be no additional reduction in time of operation if the relay current does not increase. This time is defined as the shortest period of operation. As a result, the characteristic is inverse in the beginning, and as the current increases, it tends to a fixed minimum operating time. As a result, the relay is referred to as an inverse definite minimum time relay rather than a current relay.

IV. OVER CURRENT RELAYS COORDINATION

Overcurrent relay coordination is built up throughout the system design phase using the fault current calculation. Each overcurrent relay must interact with the other protective relays on all adjacent buses to ensure that the fault is removed within a specific time frame. The importance of coordination in the design of a protection system cannot be overstated. The difficulty with overcurrent relay coordination is determining the sequence for each conceivable fault location in order to ensure that the failed section is isolated with a suitable margin of safety and without additional time delays. To accomplish an overall working time of the primary relays that is suitably minimised, a suitable time setting multiplier (TSM) and plug setting multiplier (PSM) of each relay is used to coordinate over current relays.

For large power system, protection relays and circuit breakers are the most important instruments. To separate the faulted region from the rest of the network, we require sufficient protection. Two protective equipment deployed in series are considered to be coordinated or selective if they have certain qualities that give a specific functioning sequence. Protective relay coordination aims to achieve selectivity without compromising sensitivity while also reducing fault clearance time. Because coordination methods must ensure quick, selective, and reliable relay operation to isolate the power system's faulted sections, relay coordination and type is an important part of protection system design. As shown we have many types of over current relays as shown in figure ,5 so every type of relay should be used for specific use.

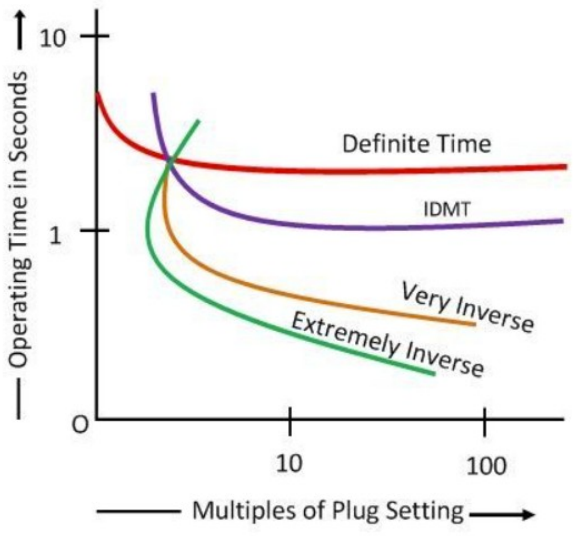

Figure 5: Over current relay graphs

V. SYSTEM DIAGRAM

Using ETAP application, a system diagram will be coordinated with over current relay. Using the methods of coordination, The system contains of feeders, grid, transformer and motor as shown in figure 6, will be coordinated to prevent faults and damage to the system. Also, using over current relay for all the grid.

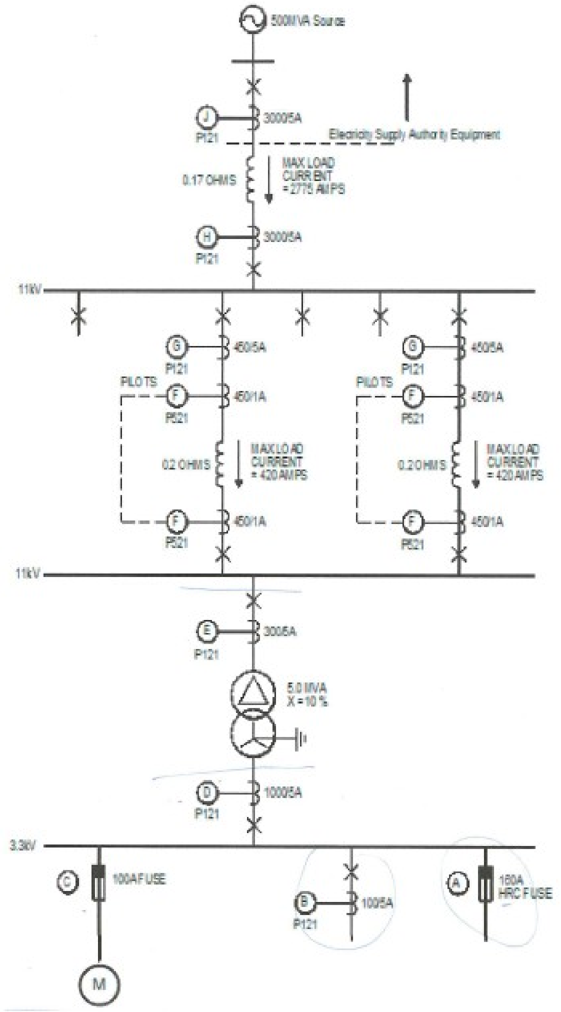

Figure 6: System diagram

1. LOAD FLOW OF THE SYSTEM

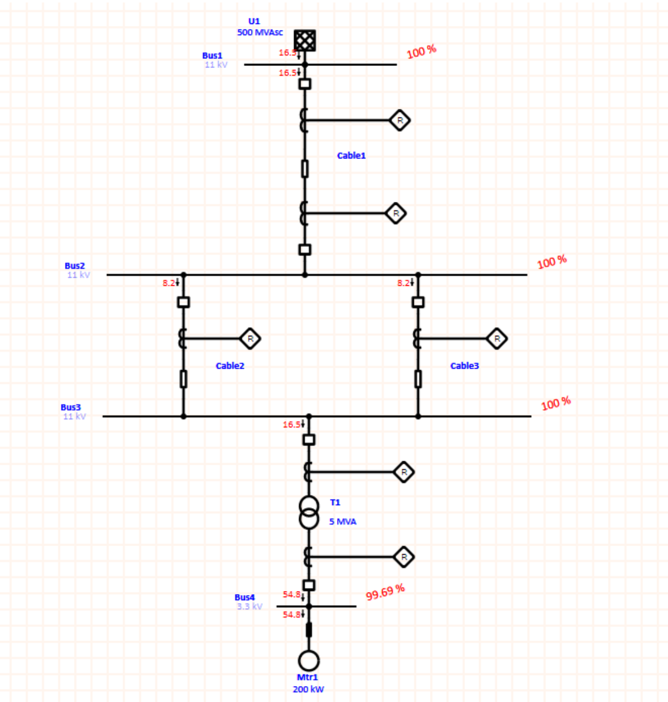

Figure 7: load flow of the system.

2. Faults on every bus

a) Bus 1

	3-Phase Fault			Line-to-Ground Fault			Line-to-Line Fault			*Line-to-Line-to-Ground		
kV	Real	Imag.	Mag.	Real	Imag.	Mag.	Real	Imag.	Mag.	Real	Imag.	Mag.
11.000	26.122	-2.697	26.260	26.119	-2.668	26.255	2.335	22.622	22.742	-15.393	-21.302	26.282

Figure 8: Bus 1 short circuit analysis

b) Bus 2

	3-Phase Fault			Line-to-Ground Fault			Line-to-Line Fault			*Line-to-Line-to-Ground		
kV	Real	Imag.	Mag.	Real	Imag.	Mag.	Real	Imag.	Mag.	Real	Imag.	Mag.
11.000	3.111	-1.019	3.273	2.518	-1.004	2.711	0.883	2.694	2.835	-0.169	3.168	3.173

Figure 9: Bus 2 short circuit analysis

c) Bus 3

	3-Phase Fault			Line-to-Ground Fault			Line-to-Line Fault			*Line-to-Line-to-Ground		
kV	Real	Imag.	Mag.	Real	Imag.	Mag.	Real	Imag.	Mag.	Real	Imag.	Mag.
11.000	3.109	-1.019	3.272	2.517	-1.004	2.709	0.882	2.693	2.834	-0.169	3.167	3.171

Figure 10: Bus 3 short circuit analysis

d) Bus 4

	3-Phase Fault			Line-to-Ground Fault			Line-to-Line Fault			*Line-to-Line-to-Ground		
kV	Real	Imag.	Mag.	Real	Imag.	Mag.	Real	Imag.	Mag.	Real	Imag.	Mag.
3.300	4.640	-5.362	7.091	4.681	-7.102	8.506	4.644	4.018	6.141	2.673	8.778	9.176

Figure 11: Bus 4 short circuit analysis

3. Relay and Fuses Graphs

VI. Calculations

Calculation is made using the values that is given in the system diagram.

Start with the relay at bus 4:

<u>Relay 6:</u>

$I_{FL6} = 54.8 \;\; Amp$

CT Ratio = 1000:5

Current tap setting:

$I_{sc} = 3080 \;\; Amp$

Check:

$$I_{sc\;\;max} = 3080\left(\frac{5}{1000}\right) = 15.4\; Amp$$

$$\left(0.1 * 54.8 * \left(\frac{5}{1000}\right) ¿\;\; 25 * 54.8 * \left(\frac{5}{1000}\right)\right)$$

$0.0274 \;¿\; 6.85$

Take current tap setting =1

$$I_{sc\,max\;\;per\;\;unit\;\;at\;\;next\;\;bus} = 9.176 * 10^{3} * \frac{\frac{5}{1000}}{1} = 15.4 \;\; p.u$$

Choose lowest TDS =0.025

T_{R6} =0.065 sec

<u>Relay 5:</u>

$I_{FL5} = 16.5 \;\; Amp$

$I_{sc} = 924 \;\; Amp$

CT Ratio = 300:5

Current tap setting:

Check

$$I_{sc\;\;max} = 924\left(\frac{5}{300}\right) = 15.4 \;\; Amp$$

$$\left(0.1 * 16.5 * \left(\frac{5}{300}\right) ¿\;\; 25 * 16.5 * \left(\frac{5}{300}\right)\right)$$

$0.0275 \;¿\; 6.875$

Take current tap setting =1

$$I_{sc\,max\,pu\;\;at\;\;next\;\;bus} = 3.272 * 10^{3} * \frac{\frac{5}{300}}{1} = 15.4 \;\;\; p.u$$

$$T_{R5} = T_{R6} + 0.4 = 0.065 + 0.4 = 0.465 \;\; sec$$

$\therefore TDS_{R5} = 0.1$ sec

<u>Relays 3 and 4:</u>

(since relay 3 and 4 are on a parallel feeder, so they will have the same coordination value)

$I_{FL\;\;3,4} = 8.2 \;\; Amp$

$I_{sc} = 462\ Amp$
CT Ratio = 450:5
Current tap setting:
Check

$$I_{sc\ \ max} = 462\left(\frac{5}{450}\right) = 5.133\ \ Amp$$

$\left(0.1*8.2*\left(\frac{5}{450}\right)\right.$ ¿ $\left.25*8.2*\left(\frac{5}{450}\right)\right)$

0.0091 ¿ 2.278

Take current tap setting =1

$$I_{sc\,max\,pu\ \ at\ \ next\ \ bus} = 3.73*10^{3}*\frac{\frac{5}{450}}{1} = 5.133\ \ p.u$$

TDS:

$T_{R4,3} = T_{R5} + 0.4 = 0.465 + 0.4 = 0.865\ \ sec$

$TDS_{R3,4} = 0.2$ sec

Relay 2:

$I_{FL\ \ 2} = 16.5\ \ Amp$

$I_{sc} = 3052.5\ \ Amp$

CT Ratio = 3000:5
Current tap setting:
Check

$$I_{sc\ \ max} = 3052.5\ \left(\frac{5}{3000}\right) = 5.0875\ \ Amp$$

$\left(0.1*16.5*\left(\frac{5}{3000}\right)\right.$ ¿ $\left.25*16.5*\left(\frac{5}{3000}\right)\right)$

0.00275 ¿ 0.6875

Take current tap setting = 0.5

$$I_{sc\,max\,pu\ \ at\ \ next\ \ bus} = 3052.5*\frac{\frac{5}{3000}}{0.5} = 5.0875\ \ p.u$$

TDS:

$T_{R2} = T_{R3,4} + 0.4 = 0.865 + 0.4 = 1.265\ \ sec$

$\therefore TDS_{R2} = 0.225$ sec

Relay 1:

$I_{FL\ \ 1} = 16.5\ \ Amp$

$I_{sc} = 3052.5\ \ Amp$

CT Ratio = 3000:5
Current tap setting:
Check

$$I_{sc\ \ max} = 3052.5\ \left(\frac{5}{3000}\right) = 5.0875\,Amp$$

$$\left(0.1*16.5*\left(\frac{5}{3000}\right) \rightarrow 25*16.5*\left(\frac{5}{3000}\right)\right)$$

$0.00275 \rightarrow 0.6875$

Take current tap setting =0.5

$$I_{sc\,max\,pu\ \ at\ \ next\ \ bus} = 5.0875*\frac{\frac{5}{3000}}{0.5} = 5.0875\ \ p.u$$

TDS:

$T_{R2} = T_{R2} + 0.4 = 1.265 + 0.4 = 1.665\ \ sec$

$\therefore TDS_{R1} = 0.25$ sec

So, to sum up all calculations in table 1.

Relays	***TDS***	t_r
Relay 6	0.025	0.065
Relay 5	0.1	0.465
Relay 3 & 4	0.2	0.865
Relay 2	0.225	1.265
Relay 1	0.25	1.665

Table 1: Relays Coordination

VI. Conclusion

In the event of an overload or failure, an overcurrent relay is an electrical protection device that cuts off the power supply to a circuit, appliance, or machine. A manual circuit breaker or contactor and a current measuring relay that interlocks with it are typical components of these devices. If the appliance or circuit is broken or overloaded, it will start drawing current beyond its regular functioning boundaries. The current sensing relay will then trip the circuit breaker or disable the contactor, cutting off the power supply. Because many machines use a lot of current when they first turn on, most overcurrent relays have a built-in "lag" feature that allows them to tolerate a high current draw for a certain amount of time before activating. The need of protection in power systems cannot be underestimated; there are numerous relays that protect power systems from overcurrent. Instantaneous Overcurrent Relay, Inverse Time Overcurrent Relay, Definite Time Overcurrent Relay, Inverse Definite Time Overcurrent Relay, Very Inverse Definite Time Overcurrent Relay, Extremely Inverse Definite Time Overcurrent Relay, Extremely Inverse Definite Time Overcurrent Relay, Extremely Inverse Definite Time Overcurrent Relay Instantaneous Overcurrent Relay, Defined Time Overcurrent Relay Choosing the right relay depends on the application and the device that needs to be secured. To isolate the failed sections in a big power system, numerous overcurrent relays are required. These relays should be synchronised in order to appropriately clear faults in a set amount of time.

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